

ANDREW XIA

AI Engineer — LLM Systems & Evals

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Portfolio: [PORTFOLIO-URL] — interactive demo, architecture & eval report

Full working rights — Temporary Graduate visa (subclass 485)

AI engineer building LLM systems with deterministic, auditable cores. Embedded in a Sydney freight forwarder's real workflow and solo-built Verdict, an air-cargo compliance engine, in two months — LLM extraction at 96.3% strict field accuracy on a 511-field ground-truth corpus, every pre-check verdict issued by rules, not the model. The AI drafts; the rules decide.

SKILLS

AI / LLM: LLM API integration (OpenAI-compatible), JSON-mode structured extraction (Zod-validated schemas), prompt engineering & versioning, confidence calibration, heuristic → LLM cascades with graceful degradation, eval harnesses & ground-truth corpora, adversarial reliability testing, token/cost observability, multi-agent dev workflows (MCP, hooks, review subagents)

Engineering: TypeScript, JavaScript, Python, SQL, PowerShell, Java · React, Vite, Zustand, Tailwind · Node.js, Express, Drizzle, PostgreSQL (Neon), SQLite · JWT, RBAC, rate limiting

Platform & QA: Git, Vitest, Playwright, GitHub Actions CI/CD, Vercel · Azure Data Factory, Power BI, Active Directory, Intune · bilingual English/Mandarin

AI PROJECTS

Verdict — Air-Cargo Compliance Engine (solo)

Apr 2026 – Present

Built inside a Sydney freight forwarder's real workflow. Interactive demo (synthetic data), case study & eval report: [PORTFOLIO-URL]/#demo

- Architected a deterministic-core / AI-periphery system — the LLM extracts, classifies and drafts; every verdict comes from a rule engine encoding 57 rules from ISPM-15, DAFF BMSB seasonal measures, IATA DGR (65th ed.), Prohibited Imports Regulations and WCO HS, with SHA-256 rule-snapshot audit trails.
- Built the LLM layer from scratch (no SDK or agent framework): provider-abstracted client with automatic failover, JSON-mode extraction against per-document-type Zod schemas, a heuristic → LLM classification cascade with graceful degradation, and per-call token/cost auditing.
- Proved it with a custom eval harness: 96.3% strict / 99.0% semantic field accuracy over an 85-document, 511-field hand-labelled corpus (clean digital English PDFs), reproducible in one command — backed by ~1,000 CI-gated tests incl. 8 adversarial rounds.

Multi-Agent Development Harness

2026 · public artifacts

- Run a multi-agent build system — orchestrator, executor and three read-only review subagents over MCP tools and custom guard hooks — that treats the coding LLM as an untrusted worker inside five layers of defense: hooks, reviewers, pre-commit gates, acceptance scripts, CI.
- Documented a failure-mode dictionary from 9 real incidents (one: the AI fabricated its own build evidence), each mapped to the safeguard that now catches it; 1,700+ line append-only engineering log.

EXPERIENCE

IT Support Technician — EVT Hotels & Resorts (ASX-listed)

Sydney · Sep 2025 – Present

- Built PowerShell automation for user lifecycle across 10,000+ users, 70+ hotels and 150+ venues — automatic AD group/permission assignment replacing manual provisioning — plus AD data-extraction and cleaning pipelines for audit; contributing to the Microsoft Intune migration.

Community Programs Volunteer (full-time), Canada — 2024 to mid 2025: organised post-pandemic mutual-aid programs for the Chinese community; returned to Sydney 2025.

Earlier — multiple IT & engineering internships and roles; full history on LinkedIn.

EDUCATION

MEng (IT), AI & Data — UNSW Sydney · Excellence, scholarship

2022 – 2023

BSc Computer Science (Minor in Mathematics) — McGill University

2016 – 2021